

POST-RACE STRATEGIC DEBRIEF

2024 Italian Grand Prix - Autodromo Nazionale Monza

Race Date: 1 September 2024 | Report Compiled: 2 September 2024

SESSION SUMMARY

Parameter	Value
Circuit	Autodromo Nazionale Monza (5.793 km)
Laps Completed	53 / 53
Starting Position	P4
Finishing Position	P2
Fastest Lap	1:21.547 (Lap 47)
Pit Stops	2 (Lap 16 Soft>Medium, Lap 38 Medium>Hard)
Total Pit Time Loss	46.3s (23.1s + 23.2s)
Weather	Dry, 28C Air / 52C Track
Safety Cars	1 (Lap 22-25)

1. PERFORMANCE & PACE ANALYSIS

Stint 1 (Laps 1-16, Soft): Average pace 1:23.142 with degradation of +0.08s/lap. Initial undercut opportunity identified on Lap 14 but delayed to Lap 16 due to traffic in pit lane.

Stint 2 (Laps 17-38, Medium): Strongest stint of the race. Average pace 1:22.876, 0.3s faster than P1 on identical compound. Degradation contained to +0.04s/lap thanks to optimal tyre preparation on out-lap.

Stint 3 (Laps 39-53, Hard): Conservative pace management to protect P2. Average pace 1:23.544. Gap to P3 managed at >3.5s throughout.

Lap Time Comparison (Selected Laps)

Lap	Time	Compound	Tyre Wear %	Gap to P1	Delta to Avg
5	1:22.987	Soft	18.2%	+1.234s	-0.155s
10	1:23.105	Soft	34.7%	+1.892s	-0.037s
15	1:23.601	Soft	52.1%	+2.445s	+0.459s
20	1:22.634	Medium	11.3%	+1.102s	-0.242s
30	1:22.811	Medium	28.9%	+0.654s	-0.065s
38	1:23.199	Medium	47.2%	+0.203s	+0.323s
45	1:23.312	Hard	15.6%	+0.891s	-0.232s
47	1:21.547	Hard	18.1%	+0.445s	-1.997s
53	1:23.889	Hard	31.4%	+1.202s	+0.345s

2. TYRE MANAGEMENT & DEGRADATION

Tyre strategy executed as planned (Soft-Medium-Hard), with the Medium stint providing the critical race advantage.

Key Findings:

- Soft compound showed higher-than-expected front-left graining in first 5 laps (+0.12s penalty vs. simulation)
- Medium compound performed 0.15s/lap better than pre-race simulation predicted, suggesting optimal track rubbering-in during the Stint 1 window
- Hard compound degradation was linear and predictable at +0.03s/lap
- Front-left tyre temperature peaked at 108.3C during Stint 1 (threshold: 110C)
- Rear tyre temperatures were well-managed at 97-101C across all stints

3. CRITICAL STRATEGY CALLS

Decision Log

Lap	Decision	Outcome	Time Delta
14	Hold pit stop (traffic)	Correct: saved 2.1s	+2.1s gained
16	Box for Mediums (undercut)	Optimal: gained P3	+1.8s gained
22	Stay out under SC	Sub-optimal: P1 pitted free	-3.2s lost
38	Box for Hards (overcut)	Correct: P1 tyres done	+0.9s gained

Net Strategy Impact: +1.6s gained vs. baseline simulation.

The Safety Car decision on Lap 22 was the key missed opportunity. Post-analysis shows pitting under SC would have cost only 14.8s vs. the 23.2s we paid at Lap 38. This single decision is the primary gap to P1 (final gap: 1.202s).

Recommendation: Update SC response protocol to weight 'free stop' probability higher when current compound has >40% wear.

4. RISK & INCIDENT ASSESSMENT

- Safety Car (Lap 22-25): Triggered by debris at Turn 4. No contact with our car. Impact: Lost strategic advantage due to decision to stay out.
- Blue Flag Incident (Lap 31): P18 backmarker delayed our car by 1.4s through Variante Ascari. FIA race control reviewed -- no penalty issued.
- DRS Failure (Lap 41-42): Intermittent DRS activation failure on main straight. Engineering confirmed rear wing actuator reset on Lap 43. Estimated time loss: 0.6s.
- Brake Temperature Alert (Lap 48): Front-left brake temperature reached 1,042C (threshold: 1,050C). Managed via ERS harvesting increase. No performance impact.

5. FORWARD-LOOKING ACTIONS (NEXT RACE: AZERBAIJAN GP)

Action Items:

1. SC Response Protocol -- Revise decision tree to prioritize 'free stop' scenarios when compound wear exceeds 40%. Target: Reduce SC-related time loss by 60%.
2. Soft Compound Preparation -- Investigate front-left graining in opening stint. Increase formation lap tyre prep temperature by 3C to improve initial grip window.
3. DRS Reliability -- Conduct full actuator diagnostic. Preemptive replacement if cycle count exceeds 2,000 activations.
4. Baku Setup Baseline -- Apply Monza high-speed aero learnings to Baku street circuit long-straight configuration. Focus on rear downforce vs. drag trade-off for Sectors 1-2.

5. Pit Stop Practice -- Target sub-2.2s stops at Baku. Monza average was 2.35s (0.15s above target). Review left-rear gunner sequence timing.

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